

Qualification Pack



Cyber Security Associate

QP Code: NIE/ITS/Q0903

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

Contents

NIE/ITS/Q0903: Cyber Security Associate	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
NIE/ITS/N0909: Fundamentals of Network and Operating System	5
NIE/ITS/N0909: Cyber Security-	10
NIE/ITS/N0910: Fundamentals of Cyber Security	14
NIE/ITS/N0910: Cyber Forensics-	20
NIE/ITS/N0911: Cryptography and Ethical Hacking	24
NIE/ITS/N0911: Perception Management	29
NIE/ITS/N0912: Network and Infrastructure Security	33
NIE/ITS/N0912: ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES- ...	38
NIE/ITS/N0913: Vulnerability Assessment and Penetration Testing.	43
NIE/ITS/N0913: Implementation of Cybersecurity & Socialmedia Analyst	47
NIE/ITS/N0914: Cyber Forensics	51
NIE/ITS/N0914: Perception Management*	55
DGT/VSQ/N0102: Employability Skills (60 Hours)	60
Assessment Guidelines and Weightage	67
<i>Assessment Guidelines</i>	67
<i>Assessment Weightage</i>	68
Acronyms	69
Glossary	70

Qualification Pack

NIE/ITS/Q0903: Cyber Security Associate

Brief Job Description

The Cyber Security Associate, course intends to build a sound Cyber Security professional for the industry. Carrying this belief forward the concepts of vulnerability and penetration testing and cyber forensics have been introduced so as to give a perspective to the enrolled learner about the two fields. On completing the course the enrolled candidate will possess knowledge about both fields and this will lay a solid foundation for the candidate's career progression.

Personal Attributes

A Cyber Security Associate should possess strong problem-solving skills to tackle complex security issues. Attention to detail is crucial to identify vulnerabilities and threats. Effective communication skills are essential for collaborating with team members and stakeholders. A strong ethical mindset and the ability to adapt to evolving technologies and threats are also important attributes.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ITS/N0909: Cyber Security-](#)
2. [NIE/ITS/N0910: Cyber Forensics-](#)
3. [NIE/ITS/N0911: Perception Management](#)
4. [NIE/ITS/N0912: ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES-](#)
5. [NIE/ITS/N0913: Implementation of Cybersecurity & Socialmedia Analyst](#)
6. [NIE/ITS/N0914: Perception Management*](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
Country	India

Qualification Pack

NSQF Level	4.5
Credits	18
Aligned to NCO/ISCO/ISIC Code	2522.0201 Security Analyst; 3513.0200 Computer Security Specialist
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) (B.Tech/ BCA/ B.Sc. (CS/IT/Electronics/ allied area)) OR Pursuing 1st year of UG (B.Tech/ BCA/ B.Sc. and continuous education) OR Pursuing 3rd year of 3-year diploma after 10th ((CS/IT/EC/EE/ allied branches) and continuous education) OR Completed 3 year diploma after 10th OR Completed 2nd year diploma after 12th OR Pursuing 2nd year of 2-year diploma after 12th (and continuous education) OR 10th grade pass with 2 year NTC plus 1 year NAC (in IT-ITeS) OR Previous relevant Qualification of NSQF Level (Qualification of Level 3 and with minimum education as 8th Grade pass and 3 years of experience in IT-ITeS Sector.)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-4.5-IT-00352-2023-V1-NIELIT
NQR Version	1

Qualification Pack

NIE/ITS/N0909: Fundamentals of Network and Operating System

Description

Fundamentals of Network and Operating Systems encompass competencies in understanding networking fundamentals, operating system basics, network configuration and management, network services and protocols, troubleshooting, virtualization, security, documentation, and compliance within the context of network and operating system environments.

Scope

The scope covers the following :

- Networking fundamentals, configuration and management, operating system basics, virtualization, security

Elements and Performance Criteria

Computer Networking

To be competent, the user/individual on the job must be able to:

- PC1.** Knowledge about the Benefits of Networks, components of Computer Networks and also have an insight into various topologies used in networking.
- PC2.** Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.
- PC3.** In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers

OSI Reference Model & TCP/IP Protocol Suite

To be competent, the user/individual on the job must be able to:

- PC4.** Overview of OSI, TCP/IP Model, and Function of layers, learn about various types of networks.
- PC5.** Knowledge about TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP, Ports, and also Some important TCP/IP protocols
- PC6.** About IP addressing, classes, IP addressing scheme, subnetting basics, how to create subnets, subnet masks, subnetting Class C addresses, subnetting Class B addresses, and VLSM include: efficiency, scalability, flexibility, and security.

TCP/IP command line Troubleshooting utilities

To be competent, the user/individual on the job must be able to:

- PC7.**
 - Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP
 - Troubleshooting utilities
- PC8.** Troubleshooting IP Addressing

Overview and Exploring Windows OS

To be competent, the user/individual on the job must be able to:

- PC9.** In-depth idea about the Windows operating system and its File System
- PC10.** Understand Basic Windows and its Control Panel
- PC11.** Understand various management service roles and get acquainted with the concept of remote desktop management.

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PC12. Students will be able to install the Windows Operating system

Overview and Exploring Linux OS

To be competent, the user/individual on the job must be able to:

PC13. The in-depth idea about the Linux operating system and its File System

PC14. Working on Command Line Utilities, Managing Users and Groups, Configuring Permissions, Read, Write, and Execute

PC15. Preparing the File System for ACLs, Configuring Administrator Tasks with sudo

PC16. Working with Links: Working with Symbolic Links, Working with Hard Links

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Computer Networking: Understanding the fundamentals of computer networking, including how data is transmitted, network topologies, common network protocols, and the importance of networks in modern computing.

KU2. OSI Reference Model & TCP/IP Protocol Suite: Mastery of the OSI reference model, which breaks down network communication into seven layers, and the TCP/IP protocol suite, which is used in the internet and real-world networking scenarios, allowing for a deep understanding of how data is encapsulated and transmitted across networks.

KU3. TCP/IP Command Line Troubleshooting Utilities: Proficiency in using command-line tools such as ping, traceroute, and ipconfig to diagnose and troubleshoot network-related issues, enhancing the ability to resolve connectivity and communication problems effectively.

KU4. Overview and Exploring Windows OS: Familiarity with the Windows operating system, including its interface, file system, user management, and basic administrative tasks, enabling students to navigate and operate within a Windows environment.

KU5. Overview and Exploring Linux OS: Competence in the Linux operating system, understanding its structure, file system, user and group management, and basic system administration tasks, allowing students to effectively use and administer Linux-based systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. A strong foundation in computer networking, OSI and TCP/IP models, command-line troubleshooting, and proficiency in Windows and Linux operating systems, providing a comprehensive understanding of core IT concepts.

GS2. A valuable set of general skills, including critical thinking, effective communication, time management, adaptability, and attention to detail, making them well-rounded professionals in the IT field.

GS3. Professional skills in network configuration, protocol understanding, operating system proficiency, security awareness, problem-solving, documentation, and user and group management, equipping them for roles in network administration, system administration, IT support, and providing a platform for continuous learning and career growth in the technology industry.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Computer Networking</i>	6	4	1	-
PC1. Knowledge about the Benefits of Networks, components of Computer Networks and also have an insight into various topologies used in networking.	-	-	-	-
PC2. Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.	-	-	-	-
PC3. In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers	-	-	-	-
<i>OSI Reference Model & TCP/IP Protocol Suite</i>	6	4	1	-
PC4. Overview of OSI, TCP/IP Model, and Function of layers, learn about various types of networks.	-	-	-	-
PC5. Knowledge about TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP, Ports, and also Some important TCP/IP protocols	-	-	-	-
PC6. About IP addressing, classes, IP addressing scheme, subnetting basics, how to create subnets, subnet masks, subnetting Class C addresses, subnetting Class B addresses, and VLSM include: efficiency, scalability, flexibility, and security.	-	-	-	-
<i>TCP/IP command line Troubleshooting utilities</i>	6	4	1	-
PC7. • Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP • Troubleshooting utilities	-	-	-	-
PC8. Troubleshooting IP Addressing	-	-	-	-
<i>Overview and Exploring Windows OS</i>	6	4	1	-
PC9. In-depth idea about the Windows operating system and its File System	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Understand Basic Windows and its Control Panel	-	-	-	-
PC11. Understand various management service roles and get acquainted with the concept of remote desktop management.	-	-	-	-
PC12. Students will be able to install the Windows Operating system	-	-	-	-
<i>Overview and Exploring Linux OS</i>	6	4	1	-
PC13. The in-depth idea about the Linux operating system and its File System	-	-	-	-
PC14. Working on Command Line Utilities, Managing Users and Groups, Configuring Permissions, Read, Write, and Execute	-	-	-	-
PC15. Preparing the File System for ACLs, Configuring Administrator Tasks with sudo	-	-	-	-
PC16. Working with Links: Working with Symbolic Links, Working with Hard Links	-	-	-	-
NOS Total	30	20	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0909
NOS Name	Fundamentals of Network and Operating System
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4.5
Credits	3.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

NIE/ITS/N0909: Cyber Security-

Description

IT aligns with National Occupational Standards (NOS) by developing core competencies in information security, database management, and network defense. Learners gain hands-on skills in configuring firewalls, proxy servers, and access controls, alongside foundational knowledge of cryptography and cyberattack mitigation. It prepares individuals to monitor, secure, and manage IT systems in accordance with industry best practices and security standards.

Scope

The scope covers the following :

- It covers foundational competencies in information security, secure server and proxy setup, firewall and ACL configuration, and basic cryptographic practices. It also develops skills for identifying cyber threats, preventing unauthorized access, and implementing defense strategies against common attacks like DoS and DDoS.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate understanding of core information security concepts, including threats, vulnerabilities, and countermeasures.
- PC2.** Successfully install and manage a database server, execute basic SQL commands, and ensure data confidentiality and integrity.
- PC3.** Configure and deploy a Linux-based proxy server with appropriate ACLs to regulate user access and internet usage.
- PC4.** Set up and manage firewall rules to permit or restrict specific services, ensuring secure network traffic flow.
- PC5.** Apply cryptographic techniques using standard industry algorithms and demonstrate encryption/decryption of data.
- PC6.** Identify and simulate common cyberattacks (e.g., DoS, DDoS, information gathering) and implement defensive measures to protect systems and data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the fundamentals of information security, including key concepts, common threats, and the importance of protecting digital assets.
- KU2.** Gain practical knowledge of firewall configuration, proxy servers, ACLs, and cryptographic techniques used to secure networks and control access.
- KU3.** Learn how to identify, analyze, and defend against cyberattacks such as hacking, DoS/DDoS, and data breaches using industry-relevant tools and methods.

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KU4. Develop skills in secure database setup, use of SQL commands, and implementing best practices to safeguard information systems against unauthorized access.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand core information security concepts, encryption methods, and common cyber threats.
- GS2.** Install and manage database servers using basic SQL commands securely.
- GS3.** Set up Linux-based proxy servers and configure firewalls and ACLs for traffic control.
- GS4.** Apply security best practices to prevent, detect, and defend against cyber-attacks.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	36	15	-	-
PC1. Demonstrate understanding of core information security concepts, including threats, vulnerabilities, and countermeasures.	36	15	-	-
PC2. Successfully install and manage a database server, execute basic SQL commands, and ensure data confidentiality and integrity.	-	-	-	-
PC3. Configure and deploy a Linux-based proxy server with appropriate ACLs to regulate user access and internet usage.	-	-	-	-
PC4. Set up and manage firewall rules to permit or restrict specific services, ensuring secure network traffic flow.	-	-	-	-
PC5. Apply cryptographic techniques using standard industry algorithms and demonstrate encryption/decryption of data.	-	-	-	-
PC6. Identify and simulate common cyberattacks (e.g., DoS, DDoS, information gathering) and implement defensive measures to protect systems and data.	-	-	-	-
NOS Total	36	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0909
NOS Name	Cyber Security-
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	5
Credits	2.4
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N0910: Fundamentals of Cyber Security

Description

Fundamentals of Cyber Security encompass competencies in core cyber security principles, regulations and policies, threat awareness, network and operating system security, access control, incident response, data protection, ethical hacking, and risk management, providing a framework for individuals to understand, implement, and adhere to essential security practices in the field of cyber security.

Scope

The scope covers the following :

- Fundamentals of Cyber Security encompasses the foundational knowledge and skills required to comprehend, implement, and maintain essential security practices, including threat awareness, network and system security, access control, and incident response, to safeguard digital assets and data from cyber threats and vulnerabilities.

Elements and Performance Criteria

Basics of Information Security

To be competent, the user/individual on the job must be able to:

- PC1.** understand Information Security, CIA Triad, Confidentiality, integrity, Availability Authentication, Authorization, and Accounting.
- PC2.** Understanding computer Ethics will be able to the differentiate between ethical and Unethical behavior
- PC3.** About Services & Techniques, Actives, Passive Threats and Exploit, etc, Authentication Authorization and Accounting (AAA)

Ethical Hacking and Basic Cryptography

To be competent, the user/individual on the job must be able to:

- PC4.** Overview of Ethical Hacking.
- PC5.** About the types of malware and its prevention.
- PC6.** About Cryptography, Plaintext, Ciphertext, Encryption, Decryption, Cipher, Symmetric, and asymmetric Cryptography, Hashing

Introduction to OS security

To be competent, the user/individual on the job must be able to:

- PC7.** About OS security and Access Control.
- PC8.** Share Security, Password Policies, User Account Policies.

Security Tools for Windows Operating System

To be competent, the user/individual on the job must be able to:

- PC9.** Overview of security tools for Windows Operating System.
- PC10.** About Virus & Threat Protection mechanisms.
- PC11.** About Firewall & Network Protection, install and configure a personal firewall, and will be able to create Inbound & Outbound Rules

Qualification Pack

Introduction to Wireless Network and Wireless Security

To be competent, the user/individual on the job must be able to:

PC12. About Wireless Networks and their Security

Android and iPhone iOS Operating systems

To be competent, the user/individual on the job must be able to:

PC13. About the security risks involved in Android OS and iOS.

Web Application Protocols & Browser Security

To be competent, the user/individual on the job must be able to:

PC14. About Web application terminologies and Browser Security.

Introduction to Social Media & its security

To be competent, the user/individual on the job must be able to:

PC15. About Social Media platforms and various cybercrimes

Digital Payment System and Security

To be competent, the user/individual on the job must be able to:

PC16. About the Cashless Economy and Payments Terminology,

PC17. Types of Payment Processing, and Security Concerns in Digital Payment System.

PC18. Various kinds of fraud in banking

Overview on Cyber Law & Cyber Cell

To be competent, the user/individual on the job must be able to:

PC19. About Cyber Crime and the background of Cyber Law.

PC20. About Cyber Cell & Cyber Fraud and helpline systems.

PC21. About registering an Online Cyber Crime, Complaint, and Registration of cybercrime complaints, Cyber Fraud Helpline

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Basics of Information Security: The importance of information security, CIA triad (confidentiality, integrity, and availability)
• Common security threats and vulnerabilities, best practices for protecting information
- KU2.** Ethical Hacking and Basic Cryptography: The principles and techniques of ethical hacking, How to identify and exploit security vulnerabilities, Basic cryptography concepts, such as encryption and decryption
- KU3.** Introduction to OS Security: The security features of popular operating systems, such as Windows, macOS, and Linux, How to harden operating systems to improve security, Common operating system security threats and vulnerabilities
- KU4.** Security Tools for Windows Operating System: How to use popular security tools to protect Windows systems, How to monitor and manage Windows security, Common Windows security vulnerabilities and how to mitigate them
- KU5.** Introduction to Wireless Network and Wireless Security: The basics of wireless networking, Common wireless security threats and vulnerabilities, Best practices for securing wireless networks

Qualification Pack

- KU6.** Introduction to the Android and iPhone iOS Operating systems: The security features of Android and iOS devices, How to harden Android and iOS devices to improve security, Common Android and iOS security threats and vulnerabilities, Android and iPhone iOS Operating systems, A more in-depth study of the security features of Android and iOS devices, How to develop secure Android and iOS apps, Advanced Android and iOS security threats and vulnerabilities
- KU7.** Web Application & Browser Security: The basics of web applications, Common web application security threats and vulnerabilities, Best practices for securing web applications, Web Application Protocols & Browser Security, The protocols used by web applications, Common web application protocol vulnerabilities, How to secure web browsers
- KU8.** Introduction to Social Media & its security: The basics of social media, Common social media security threats and vulnerabilities, Best practices for securing social media accounts
- KU9.** Digital Payment System and Security: The basics of digital payment systems, Common digital payment system security threats and vulnerabilities, Best practices for securing digital payments, Various kinds of fraud in banking, Common types of banking fraud, How to detect and prevent banking fraud
- KU10.** Overview on Cyber Law & Cyber Cell: The laws related to cyber crime in India, The role of the cyber cell in investigating and prosecuting cyber crime, Cyber Fraud Helpline, How to report cyber fraud to the authorities, What to do if you are a victim of cyber fraud

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Students studying these topics will develop general skills such as critical thinking, problem-solving, and effective communication to assess and address various cybersecurity challenges.
- GS2.** Students will gain professional skills, including the ability to apply ethical hacking techniques, secure operating systems and mobile platforms, manage digital payment systems securely, and understand the legal and law enforcement aspects of cybercrime and fraud prevention.
- GS3.** These combined general and professional skills will prepare students for careers in cybersecurity, digital forensics, or related fields, where they can safeguard digital assets, investigate cyber incidents, and contribute to the overall security of information systems.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Information Security</i>	5	3	-	-
PC1. understand Information Security, CIA Triad, Confidentiality, integrity, Availability Authentication, Authorization, and Accounting.	-	-	-	-
PC2. Understanding computer Ethics will be able to the differentiate between ethical and Unethical behavior	-	-	-	-
PC3. About Services & Techniques, Active, Passive Threats and Exploit, etc, Authentication Authorization and Accounting (AAA)	-	-	-	-
<i>Ethical Hacking and Basic Cryptography</i>	5	4	2	-
PC4. Overview of Ethical Hacking.	-	-	-	-
PC5. About the types of malware and its prevention.	-	-	-	-
PC6. About Cryptography, Plaintext, Ciphertext, Encryption, Decryption, Cipher, Symmetric, and asymmetric Cryptography, Hashing	-	-	-	-
<i>Introduction to OS security</i>	5	4	1	-
PC7. About OS security and Access Control.	-	-	-	-
PC8. Share Security, Password Policies, User Account Policies.	-	-	-	-
<i>Security Tools for Windows Operating System</i>	5	4	1	-
PC9. Overview of security tools for Windows Operating System.	-	-	-	-
PC10. About Virus & Threat Protection mechanisms.	-	-	-	-
PC11. About Firewall & Network Protection, install and configure a personal firewall, and will be able to create Inbound & Outbound Rules	-	-	-	-
<i>Introduction to Wireless Network and Wireless Security</i>	5	4	1	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. About Wireless Networks and their Security	-	-	-	-
<i>Android and iPhone iOS Operating systems</i>	5	3	-	-
PC13. About the security risks involved in Android OS and iOS.	-	-	-	-
<i>Web Application Protocols & Browser Security</i>	5	4	-	-
PC14. About Web application terminologies and Browser Security.	-	-	-	-
<i>Introduction to Social Media & its security</i>	5	3	-	-
PC15. About Social Media platforms and various cybercrimes	-	-	-	-
<i>Digital Payment System and Security</i>	5	3	-	-
PC16. About the Cashless Economy and Payments Terminology,	-	-	-	-
PC17. Types of Payment Processing, and Security Concerns in Digital Payment System.	-	-	-	-
PC18. Various kinds of fraud in banking	-	-	-	-
<i>Overview on Cyber Law & Cyber Cell</i>	5	3	-	-
PC19. About Cyber Crime and the background of Cyber Law.	-	-	-	-
PC20. About Cyber Cell & Cyber Fraud and helpline systems.	-	-	-	-
PC21. About registering an Online Cyber Crime, Complaint, and Registration of cybercrime complaints, Cyber Fraud Helpline	-	-	-	-
NOS Total	50	35	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0910
NOS Name	Fundamentals of Cyber Security
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4.5
Credits	5.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

NIE/ITS/N0910: Cyber Forensics-

Description

This module provides foundational knowledge of Indian Cyber Law and introduces core concepts of various digital forensic domains, including live, network, internet, disk, memory, and multimedia forensics. It emphasizes understanding potential digital evidence and offers practical insights into mobile forensics tools used for acquisition and analysis. The training aims to equip learners with the essential skills to assist in cybercrime investigations and digital evidence handling.

Scope

The scope covers the following :

- The scope covers foundational competencies in cyber forensics and legal frameworks. It enables learners to identify, acquire, and analyze digital evidence from various sources such as memory, disk, network, internet, multimedia, and mobile devices. The syllabus also provides a working knowledge of Indian Cyber Law, supporting legal compliance and admissibility of digital evidence in court. These competencies align with job roles like Cyber Forensic Analyst, Digital Evidence Technician, and Cyber Crime Investigator.

Elements and Performance Criteria

Indian Cyber Law

To be competent, the user/individual on the job must be able to:

PC1. Understand key provisions and applicability in cybercrime cases.

PC2. Apply legal knowledge to digital investigations ethically.

Basics of Live, Network, Internet, Disk, Memory, and Multimedia Forensics

To be competent, the user/individual on the job must be able to:

PC3. Identify and explain the purpose of each forensic type.

PC4. Demonstrate basic knowledge of tools and procedures used for data acquisition and analysis.

Potential Evidence

To be competent, the user/individual on the job must be able to:

PC5. Recognize various forms of digital evidence (e.g., files, logs, metadata).

PC6. Ensure proper documentation, handling, and preservation of evidence.

Mobile Forensics Tools

To be competent, the user/individual on the job must be able to:

PC7. Introduce commonly used tools (e.g., Cellebrite, Oxygen Forensics).

PC8. Perform basic mobile data acquisition and conduct preliminary analysis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Understand key provisions of the IT Act, cybercrime classifications, and legal procedures for handling digital evidence.
- KU2.** Identify potential evidence and apply proper methods for its collection, preservation, and documentation.
- KU3.** Get familiar with tools and techniques used to extract and analyze data from mobile devices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to evaluate digital evidence, identify patterns, and draw logical conclusions.
- GS2.** Tackling real-world cybercrime scenarios using digital forensics tools and methods.
- GS3.** Hands-on experience with mobile, memory, and network forensics tools for data acquisition and analysis.
- GS4.** Adherence to legal standards and ethical practices in digital evidence handling.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Indian Cyber Law</i>	28	6	-	-
PC1. Understand key provisions and applicability in cybercrime cases.	-	-	-	-
PC2. Apply legal knowledge to digital investigations ethically.	-	-	-	-
<i>Basics of Live, Network, Internet, Disk, Memory, and Multimedia Forensics</i>	-	-	-	-
PC3. Identify and explain the purpose of each forensic type.	-	-	-	-
PC4. Demonstrate basic knowledge of tools and procedures used for data acquisition and analysis.	-	-	-	-
<i>Potential Evidence</i>	-	-	-	-
PC5. Recognize various forms of digital evidence (e.g., files, logs, metadata).	-	-	-	-
PC6. Ensure proper documentation, handling, and preservation of evidence.	-	-	-	-
<i>Mobile Forensics Tools</i>	-	-	-	-
PC7. Introduce commonly used tools (e.g., Cellebrite, Oxygen Forensics).	-	-	-	-
PC8. Perform basic mobile data acquisition and conduct preliminary analysis.	-	-	-	-
NOS Total	28	6	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0910
NOS Name	Cyber Forensics-
Sector	Information Technology Sector
Sub-Sector	
Occupation	CYBER SECURITY
NSQF Level	5
Credits	1.3
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N0911: Cryptography and Ethical Hacking

Description

Cryptography and Ethical Hacking encompass competencies in cryptographic principles, ethical hacking practices, information security, cyber tools, network security, legal and ethical considerations, incident response, digital forensics, reporting, and a commitment to continuous learning, equipping professionals with the skills and knowledge needed for careers in cybersecurity, penetration testing, and digital forensics.

Scope

The scope covers the following :

- Cryptography and Ethical Hacking encompass a comprehensive range of competencies in information security, cryptographic techniques, ethical hacking, security tools, and legal and ethical considerations, preparing individuals to secure digital assets and perform ethical assessments of information systems and networks while adhering to professional standards.

Elements and Performance Criteria

Introduction to Cryptography and Public Key Infrastructure (PKI)

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of Cryptography and Classical and Modern Cryptography.
- PC2.** Overview of public Key Infrastructure (PKI).

Scanning and Information Gathering

To be competent, the user/individual on the job must be able to:

- PC3.** Understand Scanning and Sniffing and use the NMAP and Wireshark tools for scanning and sniffing.
- PC4.** About Detecting Network Attacks using Wireshark.
- PC5.** Overview of Information Gathering.

Open-source intelligence (OSINT) and Man in the Middle (MITM) Attacks

To be competent, the user/individual on the job must be able to:

- PC6.** Overview of Open-source intelligence (OSINT) framework.
- PC7.** Overview of MITM Attacks and their Countermeasures.

Password Cracking and Countermeasures

To be competent, the user/individual on the job must be able to:

- PC8.** Understand the types of password attacks.
- PC9.** Understand the Hash function, Hash algorithm, Password Hashes, and Password Cracking Countermeasures.
- PC10.** Understand the various types of Spoofing, Denial of Service attacks, and their countermeasures.

Steganography and E-Mail Security

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC11.** Understand the Steganography and Information Hiding Techniques.
- PC12.** Concept of Email and their protocols and understanding Email Spoofing and Phishing Techniques.
- PC13.** Understanding the concepts of Fake E-mails.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students will acquire a deep understanding of cryptography principles, encryption algorithms, and the implementation of secure communication methods, enabling them to protect data at rest and in transit effectively.
- KU2.** They will develop expertise in ethical hacking, penetration testing, and security auditing techniques, allowing them to identify vulnerabilities, assess security postures, and contribute to the proactive defense of information systems.
- KU3.** Students will gain knowledge of legal and ethical frameworks, incident response and digital forensics, network and web application security, as well as compliance requirements, equipping them with a comprehensive grasp of the ethical and technical aspects of cybersecurity and ethical hacking practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Students who engage with Cryptography and Ethical Hacking acquire general skills such as critical thinking, problem-solving, and ethical decision-making, enabling them to analyze and address complex security challenges effectively.
- GS2.** Professionally, students will develop technical skills in cryptographic techniques, ethical hacking methodologies, and vulnerability assessment, preparing them for roles in cybersecurity, digital forensics, and security auditing, while ensuring they adhere to ethical and legal standards in their practice.
- GS3.** Students cultivate communication skills that allow them to collaborate effectively with teams, communicate security findings, and contribute to the development of comprehensive security policies, enhancing their overall professional readiness in the field of information security.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Cryptography and Public Key Infrastructure (PKI)</i>	4	2	1	-
PC1. Overview of Cryptography and Classical and Modern Cryptography.	-	-	-	-
PC2. Overview of public Key Infrastructure (PKI).	-	-	-	-
<i>Scanning and Information Gathering</i>	4	3	1	-
PC3. Understand Scanning and Sniffing and use the NMAP and Wireshark tools for scanning and sniffing.	-	-	-	-
PC4. About Detecting Network Attacks using Wireshark.	-	-	-	-
PC5. Overview of Information Gathering.	-	-	-	-
<i>Open-source intelligence (OSINT) and Man in the Middle (MITM) Attacks</i>	4	3	1	-
PC6. Overview of Open-source intelligence (OSINT) framework.	-	-	-	-
PC7. Overview of MITM Attacks and their Countermeasures.	-	-	-	-
<i>Password Cracking and Countermeasures</i>	4	3	1	-
PC8. Understand the types of password attacks.	-	-	-	-
PC9. Understand the Hash function, Hash algorithm, Password Hashes, and Password Cracking Countermeasures.	-	-	-	-
PC10. Understand the various types of Spoofing, Denial of Service attacks, and their countermeasures.	-	-	-	-
<i>Steganography and E-Mail Security</i>	4	3	1	-
PC11. Understand the Steganography and Information Hiding Techniques.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Concept of Email and their protocols and understanding Email Spoofing and Phishing Techniques.	-	-	-	-
PC13. Understanding the concepts of Fake E-mails.	-	-	-	-
NOS Total	20	14	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0911
NOS Name	Cryptography and Ethical Hacking
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4.5
Credits	2.5
Version	1.0
Next Review Date	29/03/2026

Qualification Pack

NIE/ITS/N0911: Perception Management

Description

It focuses on the integration of perception management, digital media tools, and social media analytics. It equips learners with skills in web development, multimedia editing, and sentiment analysis, using modern tools and techniques. It also explores psychological warfare and propaganda, offering insights into digital influence, communication strategies, and ethical data usage.

Scope

The scope covers the following :

- It aligns with the National Occupational Standards (NOS) by developing competencies in digital content creation, social media analytics, and perception management. It equips learners with practical skills in web development, multimedia editing, and sentiment analysis using industry tools. Additionally, it fosters awareness of psychological operations, data ethics, and responsible digital communication in various professional sectors.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate understanding of the perceptual process, identify common perceptual errors, and explain the role of perception management across sectors.
- PC2.** Create and customize snippets using WordPress and apply basic web development and website optimization techniques.
- PC3.** Collect, analyze, and interpret social media data using relevant tools; perform sentiment analysis and social media listening effectively.
- PC4.** Visualize social networks and metrics accurately, while recognizing challenges in data collection, legal considerations, and ethical use.
- PC5.** Utilize multimedia tools for basic image, sound, and video editing to support digital content creation.
- PC6.** Explain psychological warfare concepts, objectives, techniques, and counter-propaganda strategies with clarity and relevance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Grasp fundamental concepts of perception, perception management, and their applications across various sectors, including understanding perceptual errors.
- KU2.** Understand the basics of web development, WordPress customization, and website optimization alongside social media analytics, including data gathering, sentiment analysis, and social media monitoring tools.

Qualification Pack

- KU3.** Comprehend social media metrics, techniques for analyzing social data, social network visualization, and challenges related to data collection, interpretation, and ethical considerations.
- KU4.** Gain insight into multimedia editing tools (image, audio, video) and foundational knowledge of psychological warfare, propaganda, and counter-propaganda strategies.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand perceptual processes, errors, and apply perception management across various sectors for effective communication.
- GS2.** Develop skills in WordPress customization, web development, multimedia editing (image, audio, video), and website optimization.
- GS3.** Analyze social media data using analytics tools, sentiment analysis, social network visualization, and apply social media listening and monitoring techniques.
- GS4.** Comprehend psychological warfare concepts, propaganda, and counter-propaganda strategies to address information influence and security challenges.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	55	29	-	-
PC1. Demonstrate understanding of the perceptual process, identify common perceptual errors, and explain the role of perception management across sectors.	55	29	-	-
PC2. Create and customize snippets using WordPress and apply basic web development and website optimization techniques.	-	-	-	-
PC3. Collect, analyze, and interpret social media data using relevant tools; perform sentiment analysis and social media listening effectively.	-	-	-	-
PC4. Visualize social networks and metrics accurately, while recognizing challenges in data collection, legal considerations, and ethical use.	-	-	-	-
PC5. Utilize multimedia tools for basic image, sound, and video editing to support digital content creation.	-	-	-	-
PC6. Explain psychological warfare concepts, objectives, techniques, and counter-propaganda strategies with clarity and relevance.	-	-	-	-
NOS Total	55	29	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0911
NOS Name	Perception Management
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	5
Credits	4.2
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N0912: Network and Infrastructure Security

Description

Network and Infrastructure Security encompass competencies in core security principles, including network and infrastructure protection, access control, incident response, encryption, risk management, and compliance, enabling individuals to effectively secure and manage digital assets and networks in various professional contexts.

Scope

The scope covers the following :

- Network and Infrastructure Security encompass competencies related to securing network and infrastructure components, including network security, operating system security, access control, incident response, vulnerability management, and adherence to security policies and compliance, providing a framework for individuals to understand, implement, and maintain essential security practices in the context of network and infrastructure protection.

Elements and Performance Criteria

Introduction to Network Security and L2/L3 Attacks

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the concept of Network Security and Network Security Protocols.
- PC2.** Understand the Attacks at Layer 3 and Layer 2
- PC3.** Understand the CAM Table and CAM Flooding Attacks.
- PC4.** Understand the Switch Port Security and Switch Port Violations.
- PC5.** Understand the concept to prevent CAM Flooding Attacks by using Switch Port Security

Network Security Control

To be competent, the user/individual on the job must be able to:

- PC6.** Understand the concept of Network Security Controls and Network Access controls
- PC7.** Understand the concept of Firewalls, DMZ, VPN, IDS, IPS, and Network Traffic Monitoring and Analysis.
- PC8.** Understand the Working of VPN, VPN Components and VPN Types.

Intrusion Detection System(IDS)

To be competent, the user/individual on the job must be able to:

- PC9.** Understand the working of IP Security, Protocols used in IPSec, the Security Architecture of IPSec, and Modes of IPSec.
- PC10.** Understand the concept of Intrusion Detection.

Understanding SOC

To be competent, the user/individual on the job must be able to:

- PC11.** Understand SOC, NOC, and the Functions of SOC.
- PC12.** Understand SIEM and Security Operations and Management

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learners will gain knowledge of fundamental network security principles, including protocols, attack types at different OSI layers, CAM table and CAM flooding attacks, and configuring switch port security.
- KU2.** Learners will understand how to prevent CAM flooding attacks, implement network access controls, and work with security measures like firewalls, VPNs, IDS/IPS, and traffic analysis.
- KU3.** Learners will grasp IP security, intrusion detection, the functions of Security Operations Centers (SOCs) and Network Operations Centers (NOCs), and the role of SIEM systems in security operations and management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Studying these topics provides learners with a robust foundation in network security, including the understanding of protocols, attack recognition, and security controls.
- GS2.** Learners gain practical skills in configuring and securing network components, effectively mitigating attacks, and utilizing critical security tools and technologies.
- GS3.** Learners develop expertise in the management of security operations, intrusion detection, and the use of SIEM systems, making them well-prepared for roles in network security and cybersecurity operations.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Network Security and L2/L3 Attacks</i>	9	1	1	-
PC1. Understand the concept of Network Security and Network Security Protocols.	-	-	-	-
PC2. Understand the Attacks at Layer 3 and Layer 2	-	-	-	-
PC3. Understand the CAM Table and CAM Flooding Attacks.	-	-	-	-
PC4. Understand the Switch Port Security and Switch Port Violations.	-	-	-	-
PC5. Understand the concept to prevent CAM Flooding Attacks by using Switch Port Security	-	-	-	-
<i>Network Security Control</i>	9	2	2	-
PC6. Understand the concept of Network Security Controls and Network Access controls	-	-	-	-
PC7. Understand the concept of Firewalls, DMZ, VPN, IDS, IPS, and Network Traffic Monitoring and Analysis.	-	-	-	-
PC8. Understand the Working of VPN, VPN Components and VPN Types.	-	-	-	-
<i>Intrusion Detection System(IDS)</i>	8	2	1	-
PC9. Understand the working of IP Security, Protocols used in IPSec, the Security Architecture of IPSec, and Modes of IPSec.	-	-	-	-
PC10. Understand the concept of Intrusion Detection.	-	-	-	-
<i>Understanding SOC</i>	8	2	1	-
PC11. Understand SOC, NOC, and the Functions of SOC.	-	-	-	-
PC12. Understand SIEM and Security Operations and Management	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	34	7	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0912
NOS Name	Network and Infrastructure Security
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4.5
Credits	1.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

NIE/ITS/N0912: ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES-

Description

This module equips learners with foundational and applied skills in Artificial Intelligence (AI) and Machine Learning (ML) using Python, with a focus on social media applications. It also covers essential system administration tasks for Windows and Linux servers, including operating system-level security features. Additionally, learners are introduced to disruptive technologies such as Big Data and Cloud Computing, enabling them to understand and navigate modern digital infrastructures effectively.

Scope

The scope covers the following :

- It equips learners with hands-on skills in managing and securing both Windows and Linux servers, essential for modern IT infrastructure. It introduces AI/ML concepts using Python with a focus on applications in social media analytics. Additionally, it offers foundational knowledge of disruptive technologies like Big Data and Cloud Computing, preparing students for roles in data-driven and cloud-based environments. Overall, it bridges system administration with emerging tech trends and intelligent automation.

Elements and Performance Criteria

AI/ML using Python in Social Media Context

To be competent, the user/individual on the job must be able to:

- PC1.** Apply Python libraries (e.g., Scikit-learn, Pandas, TensorFlow) to build basic AI/ML models.
- PC2.** Analyze social media datasets to detect trends, sentiments, and predictive patterns.
- PC3.** Integrate AI models into social media platforms for real-time decision-making or recommendation systems.

Windows Server Administration

To be competent, the user/individual on the job must be able to:

- PC4.** Configure and manage core services (Active Directory, DNS, DHCP) in a Windows Server environment.
- PC5.** Implement user and group policies to enhance system security and manage access rights.
- PC6.** Monitor and troubleshoot server performance and security using built-in tools.

Configure and secure Linux-based server systems.

To be competent, the user/individual on the job must be able to:

- PC7.** Install and configure services (e.g., SSH, Apache, firewall settings) on Linux.
- PC8.** Implement user permissions and access controls using Linux security models.
- PC9.** Apply OS-level security measures (e.g., SELinux, audit logs, iptables).

Introduction to Disruptive Technologies (Big Data & Cloud Computing)

To be competent, the user/individual on the job must be able to:

- PC10.** Describe key concepts and use cases of Big Data and Cloud Computing.

Qualification Pack

PC11. Identify major platforms (e.g., Hadoop, AWS, Azure) and their services.

PC12. Evaluate the benefits and challenges of adopting disruptive technologies in industry.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn to build machine learning models using Python and apply them to analyze social media data, such as sentiment and trends.
- KU2.** Understand how to install, configure, and manage Windows and Linux servers, including user roles and network services.
- KU3.** Learn to secure servers using built-in OS tools like firewalls, SELinux, and update management to prevent common threats.
- KU4.** Get introduced to cloud platforms and Big Data tools, exploring how they support large-scale data processing and modern IT services.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Proficient in Python-based AI/ML development and social media data analysis using popular libraries and natural language processing techniques.
- GS2.** Skilled in installing, configuring, and securing both Windows and Linux server environments, including user management, permissions, and system monitoring.
- GS3.** Experienced in applying OS-level security features such as firewalls, patch management, SELinux/AppArmor, and secure access protocols.
- GS4.** Knowledgeable about disruptive technologies, including Big Data frameworks and cloud computing models, with the ability to leverage these for scalable and innovative IT solutions.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>AI/ML using Python in Social Media Context</i>	17	9	-	-
PC1. Apply Python libraries (e.g., Scikit-learn, Pandas, TensorFlow) to build basic AI/ML models.	-	-	-	-
PC2. Analyze social media datasets to detect trends, sentiments, and predictive patterns.	-	-	-	-
PC3. Integrate AI models into social media platforms for real-time decision-making or recommendation systems.	-	-	-	-
<i>Windows Server Administration</i>	-	-	-	-
PC4. Configure and manage core services (Active Directory, DNS, DHCP) in a Windows Server environment.	-	-	-	-
PC5. Implement user and group policies to enhance system security and manage access rights.	-	-	-	-
PC6. Monitor and troubleshoot server performance and security using built-in tools.	-	-	-	-
<i>Configure and secure Linux-based server systems.</i>	-	-	-	-
PC7. Install and configure services (e.g., SSH, Apache, firewall settings) on Linux.	-	-	-	-
PC8. Implement user permissions and access controls using Linux security models.	-	-	-	-
PC9. Apply OS-level security measures (e.g., SELinux, audit logs, iptables).	-	-	-	-
<i>Introduction to Disruptive Technologies (Big Data & Cloud Computing)</i>	-	-	-	-
PC10. Describe key concepts and use cases of Big Data and Cloud Computing.	-	-	-	-
PC11. Identify major platforms (e.g., Hadoop, AWS, Azure) and their services.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Evaluate the benefits and challenges of adopting disruptive technologies in industry.	-	-	-	-
NOS Total	17	9	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0912
NOS Name	ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES-
Sector	Information Technology Sector
Sub-Sector	
Occupation	CYBER SECURITY
NSQF Level	5
Credits	1.3
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N0913: Vulnerability Assessment and Penetration Testing.

Description

Vulnerability Assessment and Penetration Testing encompass the competencies required for identifying and assessing vulnerabilities, conducting ethical penetration tests, and providing security recommendations while adhering to ethical and legal standards in the field of cybersecurity.

Scope

The scope covers the following :

- Vulnerability Assessment and Penetration Testing encompass competencies related to understanding and conducting assessments of vulnerabilities in information systems, employing legal and ethical considerations, utilizing security tools, and communicating findings to enhance security and compliance.

Elements and Performance Criteria

Vulnerability Assessment and Exploring OWASP

To be competent, the user/individual on the job must be able to:

- PC1.** Concept of Vulnerability Assessment and its Goals & Objectives.
- PC2.** Understand OWASP, Compare OWASP Top 10(2017) and OWASP Top 10 (2021) and OWASP's Top 10 vulnerabilities.
- PC3.** Understand Broken Access Control, Cryptographic Failures, and Injection

Introduction to Penetration Testing and VAPT Tools

To be competent, the user/individual on the job must be able to:

- PC4.** Concept of Insecure Design, Security Misconfiguration, Vulnerable and Outdated Components, Identification and Authentication Failures.
- PC5.** Concept of Software and Data Integrity Failures, Security Logging and Monitoring Failures.
- PC6.** About Server Side Request Forgery (SSRF), Penetration Testing and Penetration testing methodology, Types of penetration and working of Metasploit Framework.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learners studying Vulnerability Assessment and Penetration Testing will acquire knowledge and understanding of core principles in cybersecurity, including identifying and categorizing vulnerabilities, penetration testing methodologies, and legal and ethical considerations.
- KU2.** Learners will gain practical skills in using security assessment tools, assessing web application and system security, and preparing comprehensive reports
- KU3.** Learners will be well-versed in risk assessment, security best practices, and client communication, equipping them to effectively evaluate and enhance the security posture of organizations.

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learners who study Vulnerability Assessment and Penetration Testing gain general skills such as critical thinking, ethical reasoning, and effective communication, enabling them to assess and mitigate cybersecurity risks with a focus on legal and ethical considerations.
- GS2.** Professionally, learners acquire technical skills in using security tools and methodologies to identify vulnerabilities, evaluate risk, and prepare comprehensive reports for clients, equipping them to pursue careers in cybersecurity, risk assessment, and penetration testing.
- GS3.** Learners will develop a strong understanding of compliance requirements and the ability to communicate complex technical findings to stakeholders, enhancing their professional readiness in the field of cybersecurity.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Vulnerability Assessment and Exploring OWASP</i>	20	4	5	-
PC1. Concept of Vulnerability Assessment and its Goals & Objectives.	-	-	-	-
PC2. Understand OWASP, Compare OWASP Top 10(2017) and OWASP Top 10 (2021) and OWASP's Top 10 vulnerabilities.	-	-	-	-
PC3. Understand Broken Access Control, Cryptographic Failures, and Injection	-	-	-	-
<i>Introduction to Penetration Testing and VAPT Tools</i>	13	3	-	-
PC4. Concept of Insecure Design, Security Misconfiguration, Vulnerable and Outdated Components, Identification and Authentication Failures.	-	-	-	-
PC5. Concept of Software and Data Integrity Failures, Security Logging and Monitoring Failures.	-	-	-	-
PC6. About Server Side Request Forgery (SSRF), Penetration Testing and Penetration testing methodology, Types of penetration and working of Metasploit Framework.	-	-	-	-
NOS Total	33	7	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0913
NOS Name	Vulnerability Assessment and Penetration Testing.
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4.5
Credits	1.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

NIE/ITS/N0913: Implementation of Cybersecurity & Socialmedia Analyst

Description

responsible for monitoring, analyzing, and securing digital assets and social media platforms against evolving cyber threats. This role involves collecting, interpreting, and leveraging social media data to provide actionable insights while implementing cybersecurity measures to safeguard information systems.

Scope

The scope covers the following :

- Implement cybersecurity measures to protect systems and data from threats, including configuring firewalls, encryption, and access controls.
- Collect and analyze social media data using tools for sentiment analysis, social listening, and network visualization to monitor public perception and emerging risks.
- Integrate cybersecurity insights with social media intelligence to assess risks, forecast threats, and provide actionable reports to stakeholders.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Efficiently gather relevant social media data and cybersecurity logs, ensuring data accuracy, completeness, and integrity.
- PC2.** Detect and analyze potential cybersecurity threats or vulnerabilities related to social media platforms and online interactions.
- PC3.** Perform accurate sentiment analysis and social network mapping to extract meaningful insights and trends.
- PC4.** Apply appropriate cybersecurity measures such as firewalls, access controls, and encryption to safeguard data and communications.
- PC5.** Generate clear, concise reports and visualizations that communicate findings and recommendations effectively to stakeholders.
- PC6.** Ensure adherence to relevant legal, ethical, and organizational policies related to data privacy, cyber laws, and responsible social media use.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Comprehensive understanding of cybersecurity principles, including threat identification, risk management, and protection mechanisms relevant to social media platforms.
- KU2.** Knowledge of social media data structures, metrics, and analytics techniques to extract meaningful insights and monitor online behavior.
- KU3.** Familiarity with tools and methodologies for detecting, preventing, and responding to cyber threats and misinformation within social media environments.

Qualification Pack

KU4. Awareness of ethical, legal, and privacy considerations in handling social media data and implementing cybersecurity measures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to analyze and monitor social media platforms for sentiment, trends, and potential security threats using advanced data analytics tools.
- GS2.** Proficiency in identifying, assessing, and mitigating cybersecurity risks related to social media data and online communication channels.
- GS3.** Skilled in applying cybersecurity best practices, including threat detection, incident response, and implementing protective measures for digital assets.
- GS4.** Strong communication and analytical skills to generate actionable insights and reports that support decision-making in social media management and cybersecurity.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	-	-	-
PC1. Efficiently gather relevant social media data and cybersecurity logs, ensuring data accuracy, completeness, and integrity.	-	-	-	-
PC2. Detect and analyze potential cybersecurity threats or vulnerabilities related to social media platforms and online interactions.	-	-	-	-
PC3. Perform accurate sentiment analysis and social network mapping to extract meaningful insights and trends.	-	-	-	-
PC4. Apply appropriate cybersecurity measures such as firewalls, access controls, and encryption to safeguard data and communications.	-	-	-	-
PC5. Generate clear, concise reports and visualizations that communicate findings and recommendations effectively to stakeholders.	-	-	-	-
PC6. Ensure adherence to relevant legal, ethical, and organizational policies related to data privacy, cyber laws, and responsible social media use.	-	-	-	-
NOS Total	-	-	160	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0913
NOS Name	Implementation of Cybersecurity & Socialmedia Analyst
Sector	Information Technology Sector
Sub-Sector	
Occupation	CYBER SECURITY
NSQF Level	5
Credits	2
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N0914: Cyber Forensics

Description

Cyber Forensics encompasses competencies in digital evidence handling, computer and network forensics, data recovery and analysis, legal and ethical considerations, malware analysis, network traffic analysis, mobile device forensics, incident response, expert testimony, and a commitment to continuous learning, providing a framework for individuals to effectively investigate cybercrimes and cyber incidents while adhering to legal and ethical standards.

Scope

The scope covers the following :

- Cyber Forensics encompasses competencies in handling digital evidence, incident response, data recovery, forensic tools, legal and ethical considerations, network and mobile device forensics, malware analysis, file systems, expert witness testimony, documentation, and a commitment to continuous learning, providing a framework for individuals to understand, implement, and maintain effective cyber forensic practices in the context of digital investigations and legal proceedings.

Elements and Performance Criteria

Introduction to Digital Forensics

To be competent, the user/individual on the job must be able to:

- PC1.** Understand about Digital Forensics and the Objectives of Digital Forensics.
- PC2.** Understand the Elements of Digital Crime and the Process of Digital Forensics and Digital Evidence.

Digital Forensics & Acquisition

To be competent, the user/individual on the job must be able to:

- PC3.** Understand Methodology for Digital Investigators and Evidence Collection Methods.
- PC4.** Understand about Disk Imaging and Disk Cloning and the Challenges faced by Digital Forensic Investigators.
- PC5.** Understand Types of Data Acquisition, Storage Formats and Acquisition Methods

Memory Forensics

To be competent, the user/individual on the job must be able to:

- PC6.** About Data Representation, Files Examination and File Structure and File Analysis.
- PC7.** The process of Investigating Deleted Files, Formatted Disks, File Carving and File System Analysis, Memory Forensics and its Acquisition Methods.

System & Network Forensics, Logs & Reporting

To be competent, the user/individual on the job must be able to:

- PC8.** Overview of Network Forensic and logging Infrastructure
- PC9.** Using of Linux Tools for Log Analysis, Web Logs, Windows Events, and Syslog.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Learners who study the topics related to Cyber Forensics will gain a comprehensive understanding of digital evidence handling, incident response, and data recovery techniques, equipping them with the skills to effectively collect, preserve, and analyze digital evidence for legal and investigative purposes.
- KU2.** Learners will also become proficient in the use of forensic tools and methodologies, along with a strong awareness of the legal and ethical considerations involved in cyber forensics, ensuring the admissibility and integrity of evidence.
- KU3.** Learners will develop expertise in areas such as network and mobile device forensics, malware analysis, file systems, and expert witness testimony, allowing them to contribute to cybersecurity incident investigations and provide expert insights in legal proceedings.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learners who study Cyber Forensics will acquire general skills such as critical thinking, attention to detail, and effective communication, enabling them to meticulously investigate and report on digital evidence.
- GS2.** Professionally, Learners will develop expertise in digital evidence handling, incident response, and forensic analysis, preparing them for roles as cyber forensic investigators, where they apply their technical proficiency in data recovery, network forensics, and malware analysis to support legal and cybersecurity objectives.
- GS3.** Learners will possess the ethical and legal knowledge to ensure that their work adheres to legal and privacy standards and be capable of serving as expert witnesses, presenting their findings in a court of law.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Digital Forensics</i>	8	-	-	-
PC1. Understand about Digital Forensics and the Objectives of Digital Forensics.	-	-	-	-
PC2. Understand the Elements of Digital Crime and the Process of Digital Forensics and Digital Evidence.	-	-	-	-
<i>Digital Forensics & Acquisition</i>	10	3	3	-
PC3. Understand Methodology for Digital Investigators and Evidence Collection Methods.	-	-	-	-
PC4. Understand about Disk Imaging and Disk Cloning and the Challenges faced by Digital Forensic Investigators.	-	-	-	-
PC5. Understand Types of Data Acquisition, Storage Formats and Acquisition Methods	-	-	-	-
<i>Memory Forensics</i>	8	3	2	-
PC6. About Data Representation, Files Examination and File Structure and File Analysis.	-	-	-	-
PC7. The process of Investigating Deleted Files, Formatted Disks, File Carving and File System Analysis, Memory Forensics and its Acquisition Methods.	-	-	-	-
<i>System & Network Forensics, Logs & Reporting</i>	7	1	-	-
PC8. Overview of Network Forensic and logging Infrastructure	-	-	-	-
PC9. Using of Linux Tools for Log Analysis, Web Logs, Windows Events, and Syslog.	-	-	-	-
NOS Total	33	7	5	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0914
NOS Name	Cyber Forensics
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4.5
Credits	1.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

NIE/ITS/N0914: Perception Management*

Description

Cryptography and Ethical Hacking encompass competencies in cryptographic principles, ethical hacking practices, information security, cyber tools, network security, legal and ethical considerations, incident response, digital forensics, reporting, and a commitment to continuous learning, equipping professionals with the skills and knowledge needed for careers in cybersecurity, penetration testing, and digital forensics.

Scope

The scope covers the following :

- Cryptography and Ethical Hacking encompass a comprehensive range of competencies in information security, cryptographic techniques, ethical hacking, security tools, and legal and ethical considerations, preparing individuals to secure digital assets and perform ethical assessments of information systems and networks while adhering to professional standards.

Elements and Performance Criteria

Introduction to Cryptography and Public Key Infrastructure (PKI)

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of Cryptography and Classical and Modern Cryptography.
- PC2.** Overview of public Key Infrastructure (PKI).

Scanning and Information Gathering

To be competent, the user/individual on the job must be able to:

- PC3.** Understand Scanning and Sniffing and use the NMAP and Wireshark tools for scanning and sniffing.
- PC4.** About Detecting Network Attacks using Wireshark.
- PC5.** Overview of Information Gathering.

Open-source intelligence (OSINT) and Man in the Middle (MITM) Attacks

To be competent, the user/individual on the job must be able to:

- PC6.** Overview of Open-source intelligence (OSINT) framework.
- PC7.** Overview of MITM Attacks and their Countermeasures.

Password Cracking and Countermeasures

To be competent, the user/individual on the job must be able to:

- PC8.** Understand the types of password attacks.
- PC9.** Understand the Hash function, Hash algorithm, Password Hashes, and Password Cracking Countermeasures.
- PC10.** Understand the various types of Spoofing, Denial of Service attacks, and their countermeasures.

Steganography and E-Mail Security

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC11.** Understand the Steganography and Information Hiding Techniques.
- PC12.** Concept of Email and their protocols and understanding Email Spoofing and Phishing Techniques.
- PC13.** Understanding the concepts of Fake E-mails.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students will acquire a deep understanding of cryptography principles, encryption algorithms, and the implementation of secure communication methods, enabling them to protect data at rest and in transit effectively.
- KU2.** They will develop expertise in ethical hacking, penetration testing, and security auditing techniques, allowing them to identify vulnerabilities, assess security postures, and contribute to the proactive defense of information systems.
- KU3.** Students will gain knowledge of legal and ethical frameworks, incident response and digital forensics, network and web application security, as well as compliance requirements, equipping them with a comprehensive grasp of the ethical and technical aspects of cybersecurity and ethical hacking practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Students who engage with Cryptography and Ethical Hacking acquire general skills such as critical thinking, problem-solving, and ethical decision-making, enabling them to analyze and address complex security challenges effectively.
- GS2.** Professionally, students will develop technical skills in cryptographic techniques, ethical hacking methodologies, and vulnerability assessment, preparing them for roles in cybersecurity, digital forensics, and security auditing, while ensuring they adhere to ethical and legal standards in their practice.
- GS3.** Students cultivate communication skills that allow them to collaborate effectively with teams, communicate security findings, and contribute to the development of comprehensive security policies, enhancing their overall professional readiness in the field of information security.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Cryptography and Public Key Infrastructure (PKI)</i>	-	-	-	-
PC1. Overview of Cryptography and Classical and Modern Cryptography.	-	-	-	-
PC2. Overview of public Key Infrastructure (PKI).	-	-	-	-
<i>Scanning and Information Gathering</i>	-	-	-	-
PC3. Understand Scanning and Sniffing and use the NMAP and Wireshark tools for scanning and sniffing.	-	-	-	-
PC4. About Detecting Network Attacks using Wireshark.	-	-	-	-
PC5. Overview of Information Gathering.	-	-	-	-
<i>Open-source intelligence (OSINT) and Man in the Middle (MITM) Attacks</i>	-	-	-	-
PC6. Overview of Open-source intelligence (OSINT) framework.	-	-	-	-
PC7. Overview of MITM Attacks and their Countermeasures.	-	-	-	-
<i>Password Cracking and Countermeasures</i>	-	-	-	-
PC8. Understand the types of password attacks.	-	-	-	-
PC9. Understand the Hash function, Hash algorithm, Password Hashes, and Password Cracking Countermeasures.	-	-	-	-
PC10. Understand the various types of Spoofing, Denial of Service attacks, and their countermeasures.	-	-	-	-
<i>Steganography and E-Mail Security</i>	-	-	-	-
PC11. Understand the Steganography and Information Hiding Techniques.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Concept of Email and their protocols and understanding Email Spoofing and Phishing Techniques.	-	-	-	-
PC13. Understanding the concepts of Fake E-mails.	-	-	-	-
NOS Total	55	29	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0914
NOS Name	Perception Management*
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	5
Credits	4.2
Version	1.0
Next Review Date	NA

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

Qualification Pack

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N0909.Cyber Security-	36	15	0	0	51	15
NIE/ITS/N0910.Cyber Forensics-	28	6	0	0	34	25
NIE/ITS/N0911.Perception Management	55	29	0	0	84	12
NIE/ITS/N0912.ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES-	17	9	0	0	26	13
NIE/ITS/N0913.Implementation of Cybersecurity & Socialmedia Analyst	0	0	160	0	160	13
NIE/ITS/N0914.Perception Management*	55	29	0	0	84	13
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
Total	211	118	160	0	489	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.